

Product Requirements Document (PRD)

AI Trading Education Platform

Field	Detail
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1. Executive Summary

The **AI Trading Education Platform** is an AI-powered learning product designed to deliver structured, accurate, and interactive trading education to a broad audience. The platform addresses the fundamental problem of fragmented, contradictory, and often unreliable trading education available across the internet. By combining a curated knowledge base with an intelligent conversational tutor and interactive exploratory tools, the platform aims to provide learners with a trustworthy, self-paced environment in which to build market literacy.

The product is firmly positioned as an **education-first platform**. It does not provide buy/sell signals, portfolio management capabilities, broker integrations, or any form of investment advice. Its mission is to empower users with the conceptual understanding, analytical frameworks, and psychological awareness necessary to engage with financial markets responsibly.

The MVP will launch with a conversational AI tutor, structured lessons with quizzes, a daily challenge system, progress tracking, a searchable trading dictionary, and historical market data integration via Alpha Vantage. Subsequent phases will expand into interactive pattern explorers, advanced AI capabilities, and deeper gamification features.

2. Problem Statement

The current landscape of trading education is characterized by several critical shortcomings:

Problem	Description
Fragmentation	Educational content is scattered across YouTube, blogs, forums, and paid courses, making it difficult for learners to follow a coherent learning path.
Contradiction	Different sources often present conflicting strategies, definitions, and interpretations, leading to confusion among learners.
Lack of Interactivity	Most platforms offer passive content consumption (videos, articles) without hands-on practice or personalized feedback.
No Quality Assurance	There is no standardized framework for verifying the accuracy or educational quality of trading content.
Psychological Blind Spots	Most educational platforms neglect the emotional and cognitive dimensions of trading, which are critical to long-term success.

The AI Trading Education Platform addresses these gaps by offering a unified, curated, and interactive learning experience grounded in verified educational sources.

3. Target Audience

The platform serves three primary user segments, each with distinct learning needs and engagement patterns:

Segment	Profile	Key Needs
Beginners	Individuals with little to no prior experience in financial markets.	Foundational terminology, basic market mechanics, structured onboarding, and confidence-building exercises.
Intermediate Traders	Users with basic knowledge seeking to deepen their understanding of technical analysis, chart patterns, indicators, and risk management.	Advanced concept explanations, practical examples using real market data, and interactive pattern recognition tools.

Students	Academic learners (finance, economics, business) supplementing formal education with practical market knowledge.	Conceptual clarity, real-world examples, and assessment tools aligned with academic curricula.
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4. Product Scope: Modules and Content

The platform comprises twelve interconnected modules, organized into functional categories.

4.1 Core Learning Infrastructure

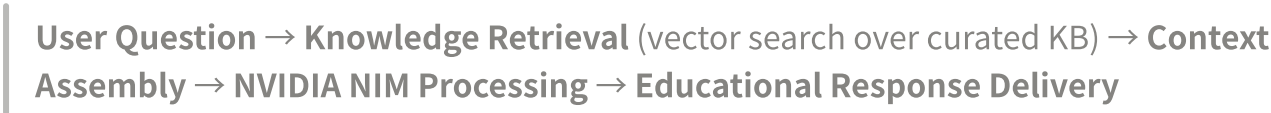
Learning Library serves as the central repository of all curated educational content. It is the authoritative knowledge base that feeds the AI Tutor's RAG pipeline and provides the backbone for structured lessons. Content is organized by topic, difficulty level, and learning path.

Learning Paths provide curated, sequential curricula that guide users through progressively complex topics. Paths are tailored to different proficiency levels (Beginner, Intermediate, Advanced) and may span topics such as "Candlestick Mastery," "Risk Management Essentials," or "Market Psychology 101."

4.2 AI Tutor

The **AI Tutor** is the platform's flagship feature: an interactive, conversational agent powered by NVIDIA NIM and grounded in Retrieval-Augmented Generation (RAG). Users can ask questions in natural language and receive contextually accurate, educational responses drawn exclusively from the platform's curated knowledge base. The AI Tutor is designed to supplement structured lessons by providing just-in-time explanations, clarifications, and deeper dives into specific topics.

The RAG flow operates as follows:



4.3 Interactive Explorers

These modules transform passive learning into active discovery through hands-on interfaces:

Explorer	Content Volume	Purpose
Candlestick Explorer	~40-45 patterns	Visual identification and practice of candlestick formations with interactive chart examples.
Chart Pattern Explorer	~25-30 patterns	Interactive study of head-and-shoulders, triangles, flags, wedges, and other classical chart formations.
Indicator Explorer	30+ indicators	Visual demonstrations of moving averages, RSI, MACD, Bollinger Bands, and other technical indicators overlaid on real market data.

4.4 Domain-Specific Education

Fundamentals covers essential financial metrics and macroeconomic indicators, including P/E Ratio, EPS, ROE, Debt analysis, Cash Flow, GDP, Inflation, and Interest Rates. Each topic includes conceptual explanations, real-world examples sourced from Alpha Vantage, and knowledge-check questions.

Market Psychology addresses the emotional and cognitive dimensions of trading. Topics include Fear, Greed, FOMO (Fear of Missing Out), Anchoring, Confirmation Bias, and Loss Aversion. This module is critical for developing the self-awareness necessary to avoid common trading pitfalls.

Risk Management provides essential instruction on capital preservation. Key topics include Position Sizing, Stop Loss strategies, Risk/Reward ratio calculations, Diversification principles, and Drawdown analysis.

4.5 Engagement and Tracking

Daily Challenge is a gamified feature that presents users with a new educational challenge each day, encouraging consistent engagement and reinforcing learning through spaced repetition. Challenges may include pattern identification, concept quizzes, or scenario-based questions.

Progress Dashboard provides a centralized view of the user's learning journey, displaying completed lessons, quiz scores, streak counts, and mastery levels across different topics.

Trading Dictionary is a comprehensive, searchable glossary of trading terminology. Each entry includes a clear definition, contextual examples, and cross-references to related concepts within the platform.

5. Minimum Viable Product (MVP)

The MVP defines the minimum set of features required to validate the product's core value proposition: an AI-powered, structured trading education experience.

#	Feature	Description
1	Conversational AI Tutor	A functional RAG-powered agent that answers educational queries using the curated knowledge base.
2	Structured Lessons	A curated subset of learning content organized into beginner and intermediate paths.
3	Interactive Quizzes	Multiple-choice and scenario-based assessments embedded within lessons.
4	Daily Challenge	A daily educational prompt or exercise to drive consistent user engagement.
5	Progress Tracking	Basic recording and display of lesson completions and quiz performance.
6	Trading Dictionary	A searchable glossary covering core trading terminology.
7	Historical Market Examples	Integration with Alpha Vantage to display real market data alongside theoretical concepts.

6. Technical Architecture

The platform is built on a modern, scalable, and developer-friendly technology stack.

Layer	Technology	Role
Frontend	Next.js + React + TailwindCSS + TypeScript	Responsive, performant SPA/SSR application with a clean, modern UI.
Backend	NestJS	Modular, enterprise-grade Node.js framework for API development and business logic.
Database	Supabase PostgreSQL + pgvector	Relational data storage with vector similarity search for RAG retrieval.
ORM	Prisma	Type-safe database client and schema migration tool.
Hosting	Vercel	Edge-optimized deployment for the Next.js frontend.
Authentication	Clerk	Secure, customizable user authentication and session management.
LLM	NVIDIA NIM	Inference engine for the AI Tutor's natural language generation.
Market Data	Alpha Vantage	Historical and intraday market data for educational examples and chart rendering.
Charting	TradingView Lightweight Charts	High-performance financial chart rendering in the browser.
Analytics	PostHog (Optional)	Product analytics for tracking user engagement, feature adoption, and retention.

7. AI System Design

7.1 Knowledge Base Construction

The AI Tutor's knowledge base is built from a carefully curated set of sources, ensuring

accuracy, originality, and regulatory compliance:

Source	Nature
Original platform content	Proprietary educational material authored specifically for the platform.
NSE educational resources	Official National Stock Exchange educational content (where permitted).
SEBI educational resources	Official Securities and Exchange Board of India educational content (where permitted).
Rewritten public concepts	Widely known financial concepts expressed in original language to avoid plagiarism.
Alpha Vantage documentation	API documentation and usage examples for market data integration.

7.2 RAG Pipeline

The Retrieval-Augmented Generation pipeline ensures that AI responses are grounded in verified educational content rather than generated freely:

1. The user submits a natural language question through the AI Tutor interface.
2. The question is embedded into a vector representation using an embedding model.
3. The embedding is used to perform similarity search over the vector-indexed knowledge base (pgvector).
4. The top-k most relevant text chunks are retrieved and assembled into a context window.
5. The context window and user question are submitted to NVIDIA NIM for response generation.
6. The generated response is post-processed through guardrail checks before delivery to the user.

7.3 Guardrails and Safety

To maintain the platform's educational integrity and regulatory compliance, the following guardrails are enforced at every layer of the AI system:

Guardrail	Enforcement Mechanism
Educational responses only	System prompt explicitly constrains the AI to educational discourse.

No investment recommendations	Keyword and intent detection triggers refusal responses for buy/sell/signal queries.
Explain concepts, psychology, and risk	Content generation is scoped to explanatory and conceptual domains.
Source attribution	Responses reference the knowledge base sources where applicable.
No hallucination tolerance	Responses are constrained to retrieved context; unsupported claims are flagged or suppressed.

8. Non-Goals and Exclusions

The following capabilities are explicitly out of scope for the current project and will not be developed in the foreseeable roadmap:

Exclusion	Rationale
Buy/Sell Signals	Providing trading signals would reposition the platform as an advisory tool, creating regulatory and liability risks.
Portfolio Management	Portfolio tracking introduces complexity around data integration, accuracy, and user expectations.
Broker Integration	Connecting to brokerage accounts requires compliance with financial regulations and introduces security concerns.
Chart Upload / Image Analysis	Vision AI, OCR, and screenshot analysis features have been deliberately removed to focus development on text-based education.

9. Content Specifications

The platform will cover a substantial breadth of trading education content, organized as follows:

Content Category	Volume	Topics
Candlestick Patterns	~40-45 patterns	Doji, Hammer, Engulfing, Morning/Evening Star, Spinning Top, etc.
Chart Patterns	~25-30 patterns	Head & Shoulders, Double Top/Bottom, Triangles, Flags, Wedges, etc.
Indicators	30+ indicators	Moving Averages, RSI, MACD, Bollinger Bands, Stochastic, ATR, etc.
Fundamentals	8 core topics	P/E, EPS, ROE, Debt, Cash Flow, GDP, Inflation, Interest Rates.
Market Psychology	6 core topics	Fear, Greed, FOMO, Anchoring, Confirmation Bias, Loss Aversion.
Risk Management	5 core topics	Position Sizing, Stop Loss, Risk/Reward, Diversification, Drawdown.

10. User Flows

10.1 Onboarding Flow

The user onboarding process is designed to be frictionless and informative:

Landing Page → **Sign Up / Login (Clerk)** → **Proficiency Selection** (Beginner / Intermediate) → **Recommended Learning Path** → **First Lesson** → **AI Tutor Introduction**

10.2 AI Tutor Interaction Flow

Users can access the AI Tutor from any point within the platform:

Any Page → **AI Tutor Chat Widget** → **User Types Question** → **RAG Processing** → **Educational Response** → **Follow-up / Related Topics**

10.3 Lesson Completion Flow

Structured lessons follow a consistent interaction pattern:

Learning Path Selection → Lesson Content → Knowledge Check Quiz → Result & Feedback → Progress Update → Next Lesson Recommendation

10.4 Daily Challenge Flow

The daily engagement mechanic operates as follows:

Dashboard Notification → Today's Challenge → Interactive Exercise → Submission → Score & Streak Update → Leaderboard (Future)

11. Success Metrics

The following key performance indicators will be used to measure the platform's success:

Metric Category	KPI	Target (MVP)
Engagement	Daily Active Users (DAU)	TBD post-launch
Engagement	Average session duration	> 8 minutes
Engagement	Daily Challenge completion rate	> 40% of active users
Learning	Lesson completion rate	> 60% per started path
Learning	Quiz pass rate (first attempt)	> 70%
AI	AI Tutor sessions per user per week	> 3
AI	AI Tutor response satisfaction	> 80% thumbs-up
Retention	7-day retention rate	> 35%
Retention	30-day retention rate	> 15%

12. Roadmap

Phase	Timeline	Focus
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Phase 1: MVP	Months 1-3	Core architecture, AI Tutor (basic RAG), structured lessons, quizzes, progress tracking, dictionary, Alpha Vantage integration.
Phase 2: Interactive Modules	Months 4-6	Candlestick Explorer, Chart Pattern Explorer, Indicator Explorer, expanded Learning Library.
Phase 3: Advanced AI & Content	Months 7-9	Enhanced RAG pipeline, expanded knowledge base, advanced psychology and risk management modules, personalized learning paths.
Phase 4: Gamification & Scale	Months 10-12	Enhanced Daily Challenge, leaderboards, community features, analytics integration (PostHog), performance optimization.

13. Assumptions and Dependencies

The following assumptions underpin the project plan and must be validated during development:

Assumption	Risk Level	Mitigation
NVIDIA NIM API availability and pricing remain stable.	Medium	Identify fallback LLM providers.
Alpha Vantage API rate limits accommodate user load.	Medium	Implement caching and rate limit management.
NSE/SEBI educational content usage is permitted under fair use.	Low	Legal review before ingestion.
User adoption aligns with MVP engagement projections.	High	Early beta testing and iterative feedback loops.

pgvector performance scales to the full knowledge base size.	Low	Benchmark and optimize indexing strategy.
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14. Open Questions

The following questions remain unresolved and should be addressed during the planning or early development phase:

1. Should the platform support multiple languages (e.g., Hindi, regional Indian languages) at MVP, or is English-only the initial scope?
2. What is the pricing model: freemium, subscription, one-time purchase, or fully free?
3. Are there specific regulatory disclosures or disclaimers required by SEBI for educational platforms of this nature?
4. Should the AI Tutor support voice input/output, or is text-only sufficient for MVP?
5. What is the target deployment timeline for the MVP?

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